

Unit 5 @ Important questions

- ①. Consider a random process $x(t)$ given by

$$x(t) = A \cos(\omega t + \phi)$$

where 'A' and ' ω ' are constants and ϕ is a uniform random variable over $[-\pi, \pi]$. Show that $x(t)$ is WSS.

- ② Consider a random process $Y(t)$ defined by:

$$Y(t) = \int_0^t x(z) dz$$

where $x(t)$ is given by:

$$x(t) = A \cos \omega t$$

where ' ω ' is constant and $A = N(0, \sigma^2)$

(a). Determine the pdf. of $Y(t)$ at $t = t_k$

(b). Is $Y(t)$ is WSS.

- ③ ~~Sta~~ For the question (1) show that the process $x(t)$ is ergodic in both the mean and the auto-correlation.

Hint: time average $\langle x(t) \rangle = \overline{x(t)} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt$

$$E[x(t)] = \int_{-\infty}^{\infty} x(t) \cdot \text{pdf} \cdot dt$$

time average of auto-correlation:-

$$\overline{R_{xx}(z)} = \langle x(t) x(t+z) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x(t+z) dt$$

$$R_{xx}(z) = E[x(t) \cdot x(t+z)]$$

$\Rightarrow$ For ergodic in both the mean and the autocorrelation

$$\langle x(t) \rangle = E[x(t)]$$

$$\overline{R_{xx}(z)} = R_{xx}(z)$$

④ Let $x(t)$ is WSS process then prove

$$(i) E\{[x(t+z) - x(t)]^2\} = 2[R_{xx}(0) - R_{xx}(z)]$$

$$(ii) R_{xx}(-z) = R_{xx}(z)$$

$$(iii) R_{xx}(0) \geq |R_{xx}(z)|$$

$$(iv) R_{xx}(0) = E[x^2(t)] = \text{Power}$$

⑤ A class of modulated random signal $y(t)$ is defined by

$$y(t) = A x(t) \cos(\omega_c t + \phi)$$

where $x(t)$ is a random message signal and $A \cos(\omega_c t + \phi)$ is the carrier. The random ~~process~~ message signal $x(t)$ is a zero-mean stationary random process with auto-correlation $R_{xx}(z)$ and power spectrum $S_{xx}(\omega)$. The carrier amplitude 'A' and the frequency ω_c are const, and phase ϕ is a random variable uniformly distributed over $[0, 2\pi]$. Assuming that $x(t)$ and ϕ are independent. Find mean, auto-correlation and power spectrum of $y(t)$.

⑥ Let $x(t)$ and $y(t)$ be both zero-mean and WSS random process. Consider the random process $z(t)$ defined by

$$z(t) = x(t) + y(t)$$

(a). Determine the auto-correlation and the power spectrum of $z(t)$, if $x(t)$ & $y(t)$ are joint WSS.

(b) repeat part (a) if $x(t)$ and $y(t)$ are orthogonal

(c) show that if $x(t)$ and $y(t)$ are orthogonal then, the mean square of $z(t)$ is equal to the sum of the mean squares of $x(t)$ and $y(t)$

⑦ Two random processes $x(t)$ and $y(t)$ are given by:

$$x(t) = A \cos(\omega t + \phi)$$

$$y(t) = A \sin(\omega t + \phi)$$

where 'A' and ' ω ' are constants and ϕ is a uniform random variable over $[0, 2\pi]$. Find the cross-correlation of $x(t)$ and $y(t)$.

⑧ A WSS random process $x(t)$ with auto-correlation

$$R_{xx}(\tau) = A e^{-a|\tau|}$$

where 'A' and 'a' are real positive constants, applied to the input of an LTI system with impulse response

$$h(t) = e^{-bt} u(t)$$

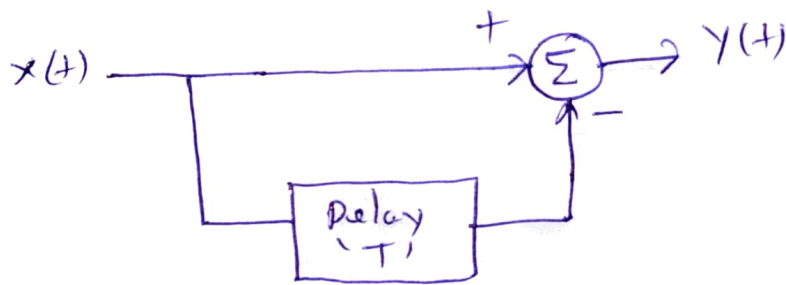
where 'b' is a real positive constant. Find the auto-correlation of the output $y(t)$ of the system.

(9) Suppose that a WSS random process $X(t)$ with power spectrum $S_{XX}(\omega)$ is the input to the filter shown in figure(1) below. Find the power spectrum of the output process $Y(t)$.

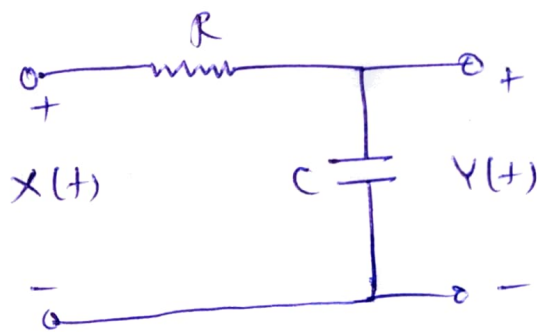
(10) The input $X(t)$ to the RC filter shown in figure(2) below is a white noise process

(a) determine the PSD of $Y(t)$

(b) determine $R_{YY}(z)$ and $E[Y^2(t)]$



Fig(1)



Fig(2)